Usman Anwar

Cambridge, United Kingdom | usmananwar391@gmail.com | uzman-anwar.github.io | linkedin.com/in/uzman-anwar
github.com/uzman-anwar | scholar.google.com/citations?user=UBAfLyQAAAAJ

Education

University of Cambridge, PhD in Engineering (Machine Learning) – Cambridge, UK — Oct 2022 – Sept 2026
- Thesis: Monitorability of Large Language Models
- Awards: Open Phil AI Fellowship and Future of Life Fellowship in AI Existential Safety

Information Technology University, MS in Data Science – Lahore, Pakistan — Sept 2019 – Aug 2021
- Rank: 2nd, Dean's Honours List
- Thesis: Inverse Constrained Reinforcement Learning (published at ICML)

University of Engineering and Technology, BS in Electrical Engineering – Lahore, Pakistan — Aug 2015 – May 2019
- Undergraduate Thesis: Single Channel Acoustic Source Separation and Speech Enhancement

Work Experience

Safety Fellow, OpenAI — Sept 2026 – Feb 2027

Astra Research Fellow, Coefficient Giving – London, United Kingdom — Jan 2026 – Apr 2026

Research Analyst, Cooperative AI Foundation – Exeter, United Kingdom — Sept 2025 – Dec 2025

Technical Research Lead, ERA Fellowship – Cambridge, United Kingdom — Apr 2025 – Sept 2025

AI Research Intern, Qualcomm AI Research – Amsterdam, Netherlands — May 2025 – Sept 2025

Senior Machine Learning Engineer, NetSol Technologies (Scientific Computing Department) – Lahore, Pakistan — Feb 2020 – Mar 2022

Awards and Grants

- *Open Phil AI Fellowship*, Open Philanthropy Foundation (2022 – 2026). Details
- *Vitalik Buterin PhD Fellowship*, Future of Life Institute (2022 – 2026).
- *Encoded Reasoning in Language Models.* UK AI Security Institute, £80,000. With D. Krueger.
- *Encoded Reasoning in Language Models.* Coefficient Giving, $50,000. With D. Krueger.
- *Graduate Student Fellowship*, ITU, Lahore.
- *Merit Scholarship*, ITU, Lahore.

Selected Publications

** denotes equal contribution. † denotes advisory role. Full list on Google Scholar.*

Preprints

Training LLMs to Verbalize Evaluation Awareness — 2026
U. Anwar, Sahar Abdelnabi, David Krueger
arxiv.org/abs/2609.36316

Monitorability Is Trainable: Post-Training for Chain-of-Thought Monitorability — 2026
U. Anwar*, Tim Bakker*, Christos Louizos, David Krueger
uzman-anwar.github.io/papers/post-training-monitorability/paper.pdf

A Decision-Theoretic Formalisation of Steganography With Applications to LLM Monitoring 2026
U. Anwar*, Julianna Piskorz*, David D. Baek, David Africa, Jim Weatherall, Max Tegmark, Christian Schroeder de Witt, Mihaela van der Schaar, David Krueger
arxiv.org/abs/2602.23163 (arXiv:2602.23163)

Analyzing and Improving Chain-of-Thought Monitorability Through Information Theory 2026
U. Anwar*, Tim Bakker*, Dana Kianfar, Cristina Pinneri, Christos Louizos
arxiv.org/abs/2602.18297 (arXiv:2602.18297)

Conferences and Journals

Understanding In-Context Learning of Linear Models in Transformers Through an Adversarial Lens 2025
U. Anwar, Johannes von Oswald, Louis Kirsch, David Krueger, Spencer Frei
openreview.net/forum?id=CtMXJxO7SJ (Transactions on Machine Learning Research (Featured Certification))

Interpreting Emergent Planning in Model-Free Reinforcement Learning 2025
Thomas Bush, **U. Anwar*†**, Stephen Chung*, Adrià Garriga-Alonso, David Krueger
openreview.net/forum?id=DzGe40glxs (ICLR (Oral))

Mitigating Goal Misgeneralization via Minimax Regret 2025
Matthew Farrugia-Roberts*, Karim Ahmed Abdel Sadek*, Hannah Erlebach, Christian Schroeder de Witt, David Krueger, **U. Anwar†**, Michael D. Dennis
openreview.net/forum?id=okCqx1RdIw (Reinforcement Learning Conference)

Reward Model Ensembles Help Mitigate Overoptimization 2024
Thomas Coste, **U. Anwar†**, Robert Kirk, David Krueger
openreview.net/forum?id=dcjtMYkpXx (ICLR)

Foundational Challenges in Assuring Alignment and Safety of Large Language Models 2024
U. Anwar, Abulhair Saparov, Javier Rando, Daniel Paleka, and others
openreview.net/forum?id=oVTkOs8Pka (Transactions on Machine Learning Research (Survey Certification))

The Reality of AI and Biorisk 2025
Aidan Peppin, Anka Reuel, Stephen Casper, Elliot Jones, Andrew Strait, **U. Anwar**, and others
arxiv.org/abs/2412.01946 (ACM Conference on Fairness, Accountability, and Transparency (FAccT))

Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback 2023
Stephen Casper, Xander Davies, **U. Anwar**, and others
openreview.net/forum?id=bx24KpJ4Eb (Transactions on Machine Learning Research (Survey Certification, Featured Certification))

Bayesian Methods for Constraint Inference in Reinforcement Learning 2022
Dimitris Papadimitriou, **U. Anwar**, Daniel S. Brown
openreview.net/forum?id=oRjk5V9eDp (Transactions on Machine Learning Research)

Inverse Constrained Reinforcement Learning 2021
U. Anwar*, Shehryar Malik*, Alireza Aghasi, Ali Ahmed
proceedings.mlr.press/v139/malik21a.html (ICML (Spotlight))

Workshops

IDs for AI Systems 2024
Alan Chan, Noam Kolt, Peter Wills, **U. Anwar**, Christian Schroeder de Witt, Nitarshan Rajkumar, Lewis Hammond, David Krueger, Lennart Heim, Markus Anderljung
openreview.net/forum?id=0rQGtt50rZ (NeurIPS Workshop on Regulatable ML)

Learning to Forget using Diffusion Hypernetworks 2024
Jose Miguel Lara Rangel, Stefan Schoepf, Jack Foster, David Krueger, **U. Anwar†**
openreview.net/forum?id=vvRQx80XqX (NeurIPS Workshop on New Frontiers in Adversarial Machine Learning)

Comparing Bottom-Up and Top-Down Steering Approaches on In-Context Learning Tasks 2024
Madeline Brumley, Joe Kwon, David Krueger, Dmitrii Krasheninnikov, **U. Anwar†**
openreview.net/forum?id=VMiWNaWVJ5 (MINT: Foundation Model Interventions Workshop)

Noisy ZSC: Breaking The Common Knowledge Assumption In Zero-Shot Coordination Games 2023
U. Anwar*, Ashish Pandian*, Jia Wan, David Krueger, Jakob Nicolaus Foerster
arxiv.org/abs/2411.04976 (NeurIPS Workshop on Open Ended Learning)

Public Writing

- **Paraphrasing Is (At Best) a Partial Defence Against Steganography in LLMs** — with Rob Krzyzanowski. LessWrong, May 2026.
- **What Will AI Firms Look Like?** — Unaligned Thoughts, February 2026.
- **Three Archetypes of AI Agents — or Why Claude (Code) and Claudius Are Different Kinds of Agents** — Unaligned Thoughts, February 2026.
- **EU Regulations Are Not Ready for Multi-Agent AI Incidents** — with Natàlia Fernández Ashman and Marta Bieńkiewicz. Tech Policy Press, January 2026.

Teaching Experience

Teaching Assistant — Discrete Mathematics, Information Technology University (Department of Computer Science) – Lahore, Pakistan — Sept 2019 – Jan 2020

Selected Talks

- **A Decision-Theoretic Formalisation of Steganography With Applications to LLM Monitoring** — FAR AI Alignment Workshop (video), ML Collective Deep Learning: Classics and Trends reading group (August 2026).
- **Analyzing and Improving Chain-of-Thought Monitorability Through Information Theory** — ERA AI Safety Fellowship, Cambridge AI Safety Hub.
- **Foundational Challenges in LLM Alignment and Safety** — SERI MATS, ACM Summer School on Responsible & Safe AI, GovAI, Chalmers AI Ethics Seminar and others.
- **Reward Modelling for AGI Safety** — Future of Life Seminar on AI Safety, October 2022.

Professional Activities and Services

- **Lead Organizer**: NeurIPS Workshop on Socially Responsible Language Modelling Research (SoLaR), 2023 & 2024.
- **Peer Reviewer**: ICML 2023, NeurIPS 2023, ICLR 2025.
- **Grant Reviewer**: Vitalik Buterin Fellowship for AI Existential Safety (2023).
- **Mentor**: GradAppLab (2022 – Present).

Mentoring and Supervisions

- Karim Abdul Sadek — *Mitigating Goal Misgeneralization via Minimax Regret* at University of Cambridge (accepted at RLC 2025).
- Thomas Bush — *Interpreting Emergent Planning in Model-Free Reinforcement Learning* at University of Cambridge (accepted at ICLR 2025, Oral).
- Miguel Lara — *Learning to Forget Using Diffusion Hypernetworks* at University of Cambridge (accepted at NeurIPS 2024 Workshop on New Frontiers in Adversarial ML).
- Thomas Coste — *Conservative Agency for Safe Optimization of Learned Reward Models* at University of Cambridge (accepted at ICLR 2024).
- Yawen Duan — *Red Teaming (Learned) Reward Models* at University of Cambridge.
- Jason Brown — *Likert Scale Feedback to Improve Robustness of Learning From Human Preferences* at University of Cambridge.
- Abdul Rehman & Arslan Malik — *Privacy Preserving Recommendation System* at ITU, Pakistan.

Volunteer Work

Managing Director & Co-Founder, Spectra Magazine – Lahore, Pakistan Apr 2017 – May 2020

- Spectra Magazine is a student-powered online magazine aiming to enhance public understanding of science and shape the narrative of science journalism in Pakistan.
- Under my leadership, we published more than 215 articles and mentored more than 50 high school and undergraduate students in science writing, editing and design. Read more at spectramagazine.org/about.

Non-Degree Studies

- Cooperative AI Summer School (July 2023).
- Eastern European Machine Learning School (July 2021).

Skills

Languages: Python, C, SQL, NoSQL

ML Frameworks: PyTorch, JAX, TensorFlow

Scientific Computing: NumPy, SciPy, Matplotlib